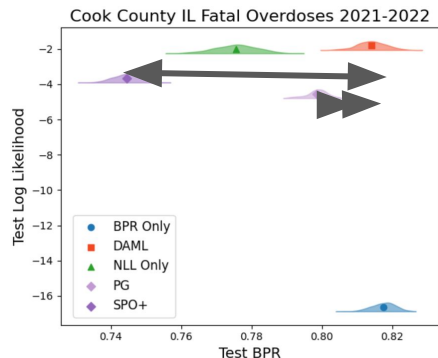
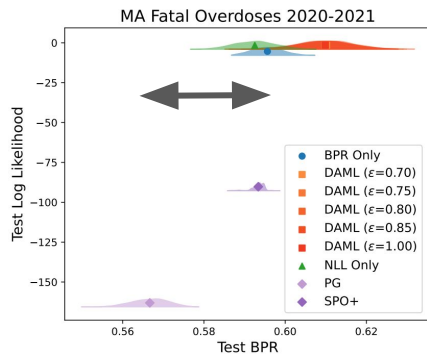
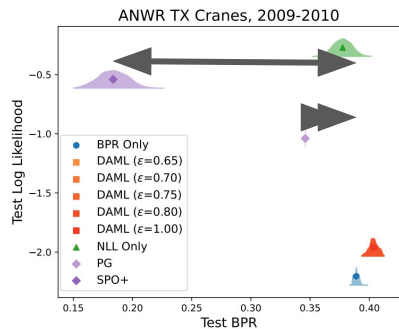


The Opportunity



Why do these methods, which are trained on a theoretically similar objective, come to such different solutions?





NEURAL INFORMATION
PROCESSING SYSTEMS

BENCHMARKING PTO AND PNO METHODS IN THE PREDICTIVE COMBINATORIAL OPTIMIZATION REGIME

Haoyu Geng, Hang Ruan, Runzhong Wang, Yang Li, Yang Wang, Lei Chen, Junchi

Table 3: Results for 7 problems on 11 methods. The relative regret (w.r.t. optimal objective %) and average runtime of each epoch are reported. “-” means non-applicable and data generation time of LODL is not counted. Best three: **red**, **orange**, **blue**.

Problem		Predictive	Discrete			Continuous		Statstical				Surrogate
		Two-stage	DFL	Blackbox	Identity	CPLayer	SPO	NCE	point-LTR	pair-LTR	list-LTR	LODL
Knapsack (Gen)	Regret (%)	6.595	11.744	24.274	31.874	24.769	6.223	13.438	6.402	7.820	6.031	6.044
	Train Time	0.101	1.208	2.195	1.351	0.368	1.173	2.065	1.846	4.861	1.536	0.344
	Test Time	0.794	1.636	1.544	1.553	1.928	1.100	1.445	1.682	3.973	1.119	0.716
Knapsack (Energy)	Regret (%)	8.745	8.353	35.705	17.156	36.402	8.407	11.932	8.236	9.022	8.083	9.567
	Train Time	0.198	1.116	1.776	1.051	0.416	2.007	4.010	2.602	10.396	3.099	0.501
	Test Time	0.342	0.760	0.752	0.745	1.825	1.069	2.054	1.406	6.751	1.699	0.344
Scheduling (Energy)	Regret (%)	1.793	6.272	6.503	5.690	-	1.505	1.663	4.548	1.540	1.551	1.786
	Train Time	0.404	51.857	110.647	56.334	-	110.362	109.645	65.744	61.048	61.873	0.582
	Test Time	13.196	26.186	26.799	26.920	-	41.448	37.904	28.590	27.100	26.945	12.055
Budget Allocation	Regret (%)	20.332	35.970	26.905	14.799	-	5.559	9.979	69.663	5.958	5.742	25.700
	Train Time	0.102	4.019	39.799	21.224	-	40.021	7.327	22.669	23.194	22.774	0.248
	Test Time	12.828	25.138	25.728	26.344	-	38.127	38.460	26.392	26.293	26.168	12.278
TopK (Cubic)	Regret (%)	0.110	1.974	13.944	13.944	-	160.408	160.408	1.149	5.072	0.193	0.172
	Train Time	0.064	0.116	0.096	0.097	-	0.126	0.393	0.379	4.653	0.679	0.197
	Test Time	0.038	0.105	0.090	0.087	-	0.125	0.890	0.832	11.625	1.571	0.034
Bipartite Matching	Regret (%)	92.963	91.364	91.988	91.868	92.007	93.327	92.622	91.035	92.285	91.831	91.113
	Train Time	0.010	25.121	0.623	0.334	17.179	1.895	3.467	0.659	1.501	1.531	0.051
	Test Time	0.262	7.725	0.446	0.457	6.643	1.160	1.758	0.886	1.055	1.044	0.239
Portfolio	Regret	0.243	0.380	0.286	0.280	0.309	0.245	0.367	0.214	0.255	0.249	0.160
	Train Time	0.204	0.754	3.762	2.018	1.121	3.446	11.082	3.382	16.383	3.564	0.198
	Test Time	1.187	3.547	2.251	3.115	3.504	3.121	8.071	3.829	17.087	3.919	1.182

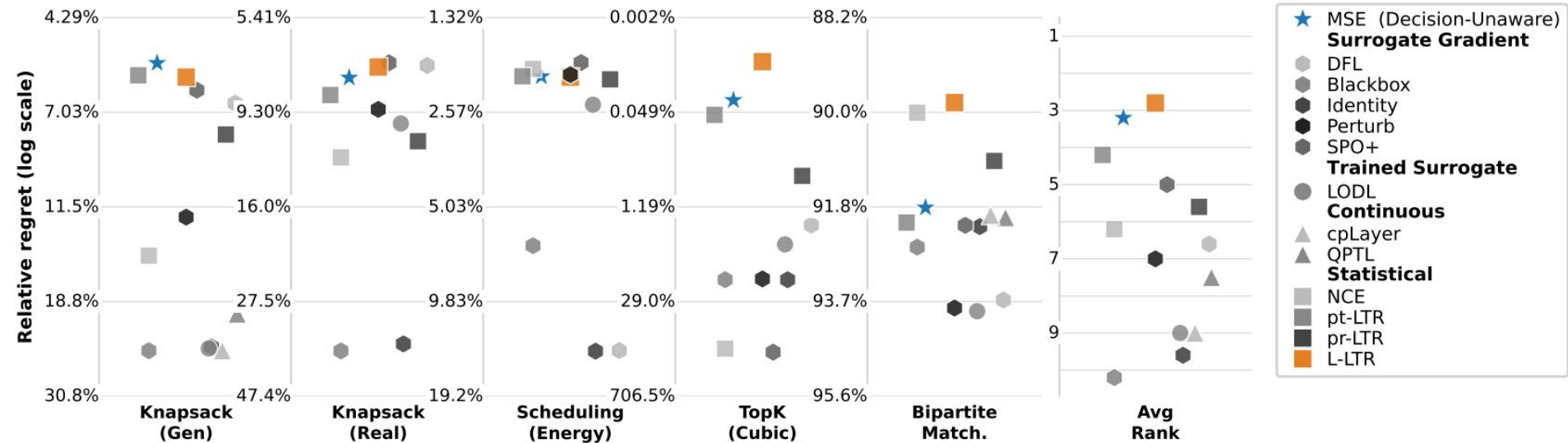
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“However, no silver bullet is found for specific design choices...”

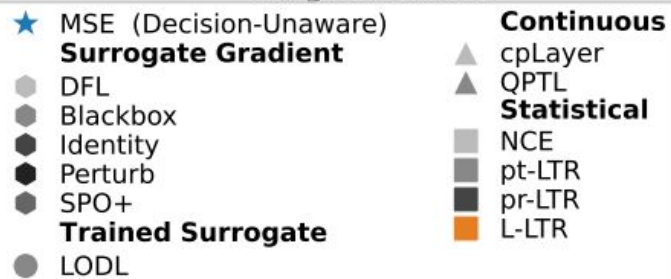
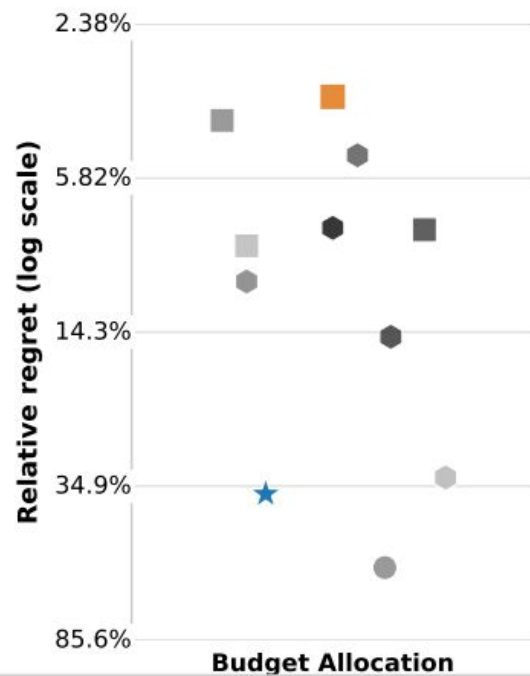
- Original: 5 Learning Rates
- Rerun:
 - Phase 1: 3 Learning Rates, 2 Batch Sizes
 - Phase 2: 4 new hyperparameter settings for methods with their own hyperparameters

Benchmark re-run — Relative regret % (best ↑)



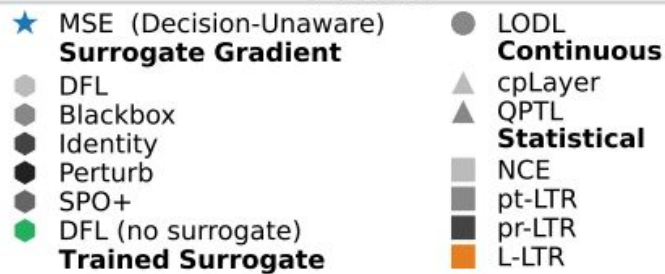
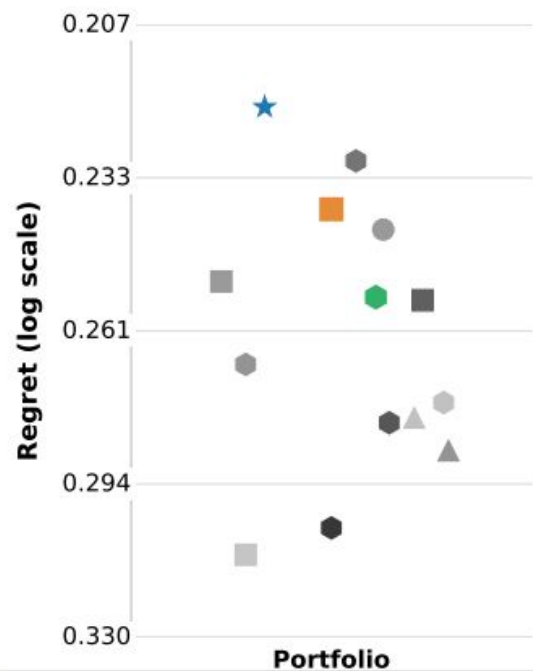
Benchmark re-run — Budget Allocation

Relative regret % (best ↑)

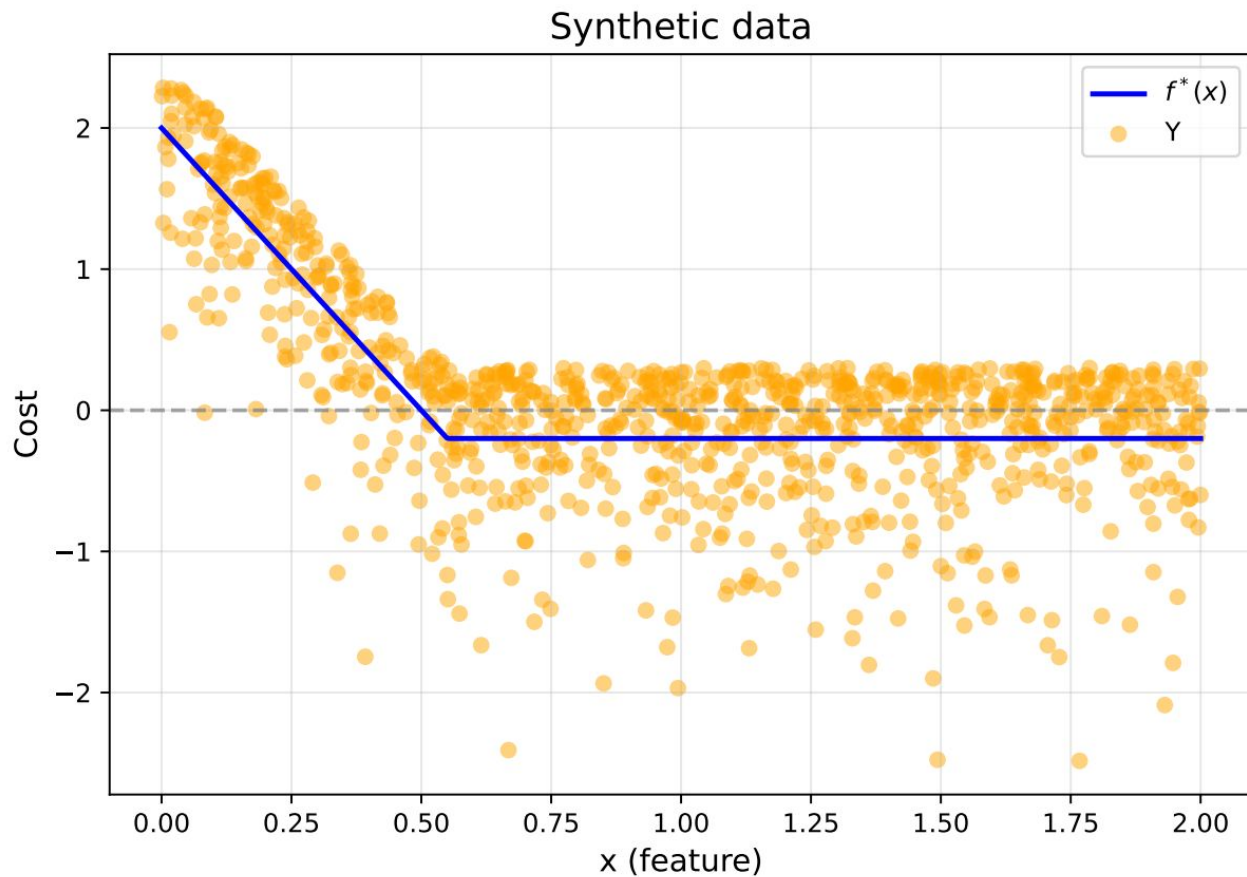


Benchmark re-run — Portfolio

Absolute regret (best ↑)



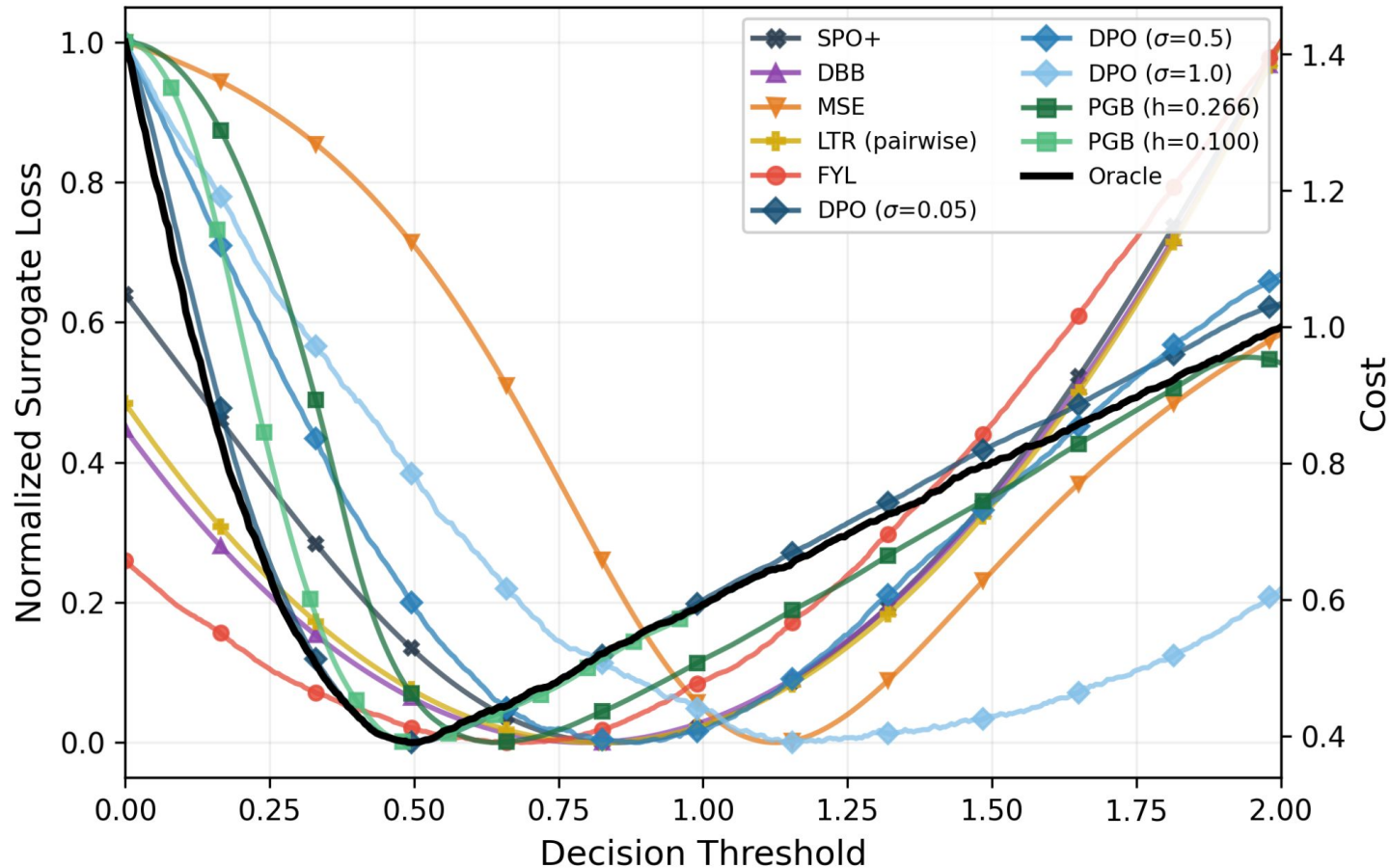
How to make a task where decision-aware methods are necessary



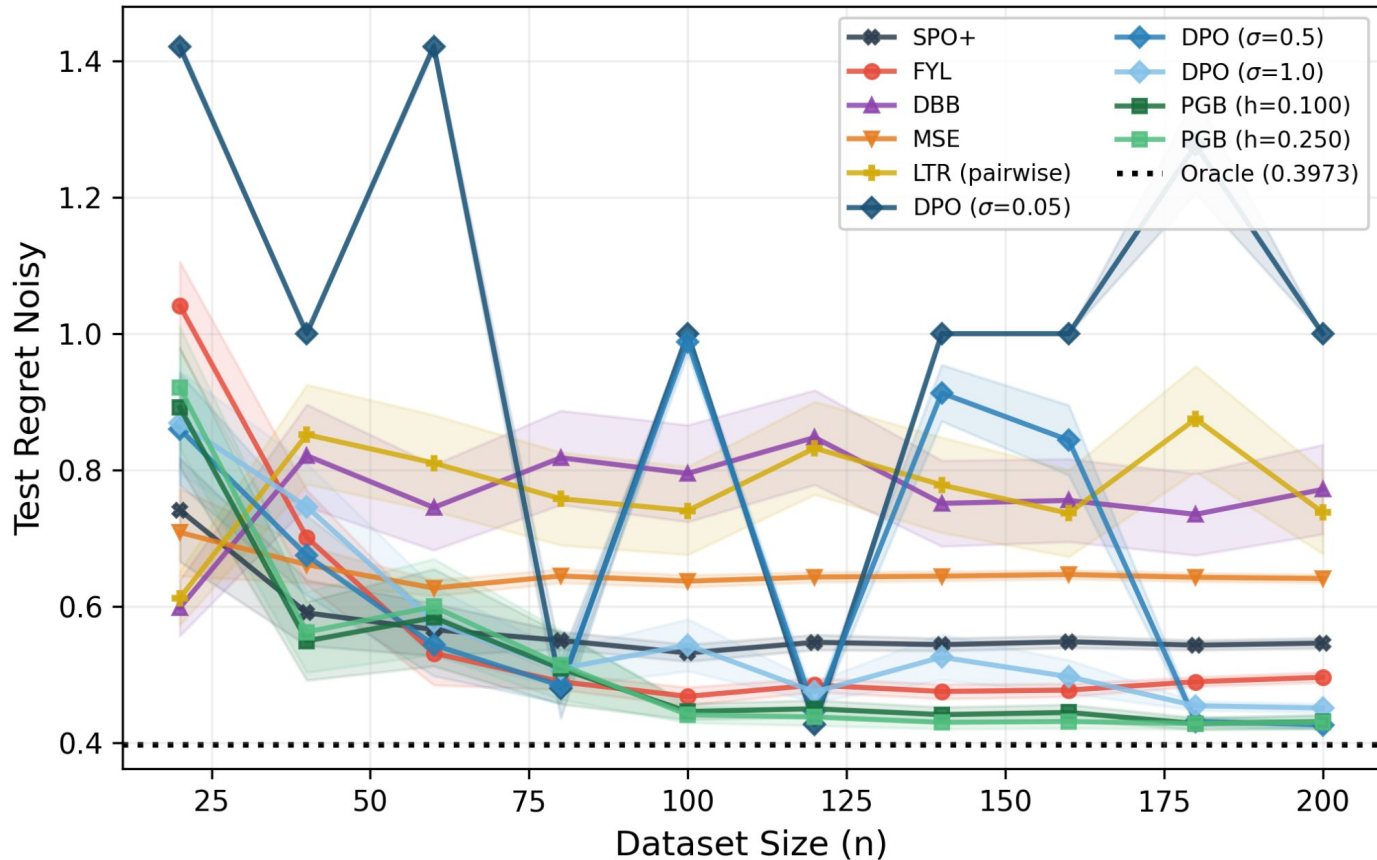
When evaluating decision-aware learning methods, there are two dimensions to consider:

1. What level of regret does a method prefer?
2. What level of regret is a method capable of learning?

What level of regret does a method prefer?



What level of regret is a method capable of reaching?



What level of regret is a method capable of reaching?

